Permutations — Problem Set

Warm-up

1. How many ways can 5 books be arranged on a shelf?
2. Find the number of arrangements of the letters in the word **LOGIC**.
3. How many ways can 6 different books be arranged on a shelf?
4. In how many ways can the letters of the word **CIDER** be rearranged?
5. How many ways can 5 people be arranged in a line if a specific person must stand at the front?
6. Find the number of distinct arrangements of the letters in the word **APPLE**.
7. In how many ways can 4 people be seated around a circular table?
8. How many 3-digit numbers can be formed using the digits 1, 2, 3, 4, 5 if no digits are repeated?
9. If 3 people must sit together in a row of 6, how many distinct arrangements are possible?
10. How many arrangements of the word **BANANA** exist?
11. A license plate consists of 2 letters followed by 3 digits. How many plates are possible if repetition is allowed?

Skill-building

1. In how many ways can the letters of the word **KITCHEN** be arranged if the vowels must stay together?
2. Find the number of ways to arrange 8 people in a row if 2 particular people refuse to sit next to each other.
3. How many distinct arrangements of the word **MISSISSIPPI** are possible?
4. Seven people are to be seated at a round table. In how many ways can this be done if two people, A and B, must sit next to each other?
5. How many even 4-digit numbers can be formed from the digits 1, 2, 3, 4, 5, 6 without repetition?
6. In how many ways can the letters of the word **CHORUS** be arranged if the word must start and end with a consonant?
7. Five boys and five girls are to be seated in a row. How many arrangements are possible if the boys and girls must alternate?
8. In how many ways can 6 people be arranged in a line if 3 specific people must be separated from each other?

9. Five people are to be seated at a circular table. In how many ways can this be done if two particular people must sit next to each other?

10. How many ways can the letters of the word **CHORUS** be arranged if the word must start with a consonant?

Easier Exam Questions

1. (Sydney Girls 2022 Q1a)
   How many ways can 10 people be arranged in a straight line? **1**

2. (Girraween 2022 Q1)
   Naveen has got 7 friends. Number of ways in which he can invite two or more of them for a party is

   (A) 127
   (B) 120
   (C) 21
   (D) 56

3. (Girraween 2024 Q3)
   In how many ways can 7 boys be arranged in a line if Peter, Jeffery, and Kevin have to stand together at the beginning of the line?

   (A) 4!
   (B) 7!
   (C) $4! \times 3!$
   (D) $7! \times 3!$

4. (Sydney Girls 2021 Q1a)
   The number plates for scooters appear as 3 letters followed by 2 numbers such as MAX21. How many different number plates are possible? **1**

5. (Caringbah 2025 Q6c)
   How many possible distinct arrangements can be made by using the letters of the word POSSIBILITY?

   (i) with no restrictions? **1**
   (ii) If both letters S are together? **1**
   (iii) If the letter S occupies both first and last place? **1**

6. (Glenwood 2023 Q8c)
   In how many ways can the letters of the word 'COMMITTEE', taken all at a time , be arranged if:

   (i) there is no restriction? **1**
   (ii) the two E's stay together? **1**

7. (Blacktown Boys 2022 Q5)
   In how many ways can 7 boys be arranged in a line if Peter, Jeffery and Kevin have to stand together at the beginning of the line?

(A) $4!$

(B) $7!$

(C) $4! \times 3!$

(D) $7! \times 3!$

8. (Sydney Girls 2019 Q1a)
How many different arrangements of the letters of the word MATHS are possible? **1**

9. (Gosford 2022 Q4a)
Let each different arrangement of all the letters of DELETED be called a word.

i. How many different words will be possible? **1**

ii. In how many of these words will the D's be separated? **2**

10. (Sydney Girls 2020 Q2f)
Five boys and six girls are to be seated in a row. Find the number of ways this can be done if:

(i) there are no restrictions; **1**

(ii) the boys all sit together. **1**

11. (Presbyterian 2021 Q4)
How many seven-letter arrangements can be made using the letters of the word DELETED?

(A) 420

(B) 840

(C) 2520

(D) 5040

12. (Knox 2022 Q1)
How many distinct ways are there to arrange the letters of the word EMBEZZLED?

(A) 3024

(B) 8640

(C) 30 240

(D) 362 880

Harder Exam Questions

1. (Baulkham Hills 2024 Q1)
How many ways can 7 people be seated around a circular table if 2 particular people, Preksha and Chloe, do not wish to sit next to each other?

(A) 480

(B) 718

(C) 2880

(D) 3360

2. (Presbyterian 2023 Q9)
In the school playground, 2 girls and 4 boys are seated in a circle. How many arrangements are possible if the girls cannot sit next to each other?

(A) 18
(B) 24
(C) 60
(D) 72

3. (Gosford 2020 Q7)
In how many ways can a group of 6 people line up in a queue so that the oldest person is in front of the youngest person?

(A) 180
(B) 360
(C) 540
(D) 720

4. (Girraween 2023 Q4)
In how many ways can the letters of the word TROOPS be arranged if the two Os are to be separated?

(A) 60
(B) 240
(C) 300
(D) 600

5. (Girraween 2021 Q2 a,b & d)
a. In how many ways can the letters of the word DEDUCED be arranged? **2**

b. How many numbers greater than 3 000 can be formed from the digits 1, 3, 5, 7, 9 if no repetition is permitted? **2**

d. Mr and Mrs Antony (the hosts) and 6 guests sit around a circular dining table. In how many ways can they be arranged if the two hosts are separated? **2**

6. (Barker 2023 Q2)
In how many ways can the letters of the word ROAMING be arranged if the new word must end in a vowel?

(A) 720
(B) 2160
(C) 4320
(D) 5040

7. (Knox 2019 Q12c)
The letters of STREET are arranged randomly in a row. Find the probability that:

(i) the two letters E are together. **2**

(ii) the Es and Ts are together in one group. **2**

8. (Hurlstone 2024 Q2)
Eight chairs are equally spaced around a circular table. What is the number of ways four teachers, and four students can be seated at the table so that no student is directly opposite a teacher?

   (A) 36
   (B) 144
   (C) 432
   (D) 5040

9. (SCEGGS 2023 Q4)
The letters of the word REPETITION are arranged at random in a row. How many arrangements will have vowels and consonants alternating?

   (A) 1800
   (B) 3600
   (C) 14400
   (D) 28800

10. (Glenwood 2020 Q1)
In how many ways can five boys and two girls be arranged in a row if the two girls are together?

   (A) 300
   (B) 1440
   (C) 10
   (D) 10080

11. (Sydney Tech 2020 Q4)
How many ways can the letters in the word QUACKED be arranged into 7 letter words if the consonants and vowels must alternate?

   (A) $4!$
   (B) $4! \times 3$
   (C) $4! \times 3!$
   (D) $4 \times 3!$

12. (SCEGGS 2021 Q2)
Libby owns five non-identical black dresses and four non-identical white dresses. In how many ways can she arrange the dresses in a line in her cupboard if the black and white dresses alternate?

   (A) $5!4!$
   (B) $2!5!4!$
   (C) $\frac{9!}{5!4!}$
   (D) $\frac{5!4!}{2}$

13. (Knox 2021 Q12a)
Eight friends are seated in a row at the cinema. The seats on each end of the row are aisle seats.

In how many ways can they be seated if Alice and Bob insist on sitting in an aisle seat and Corinne sits the same number of seats away from Douglas as she does from Ellen? **2**